BAHRIA UNIVERSITY, LAHORE CAMPUS
DEPARTMENT OF COMPUTER SCIENCE

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Class: BSCS-5B **Topic:** UAV Fleet Disaster Management

UAV Fleet Disaster Management Using Genetic Algorithm

Technical Design & Empirical Evaluation Report

Abstract— Unmanned Aerial Vehicle (UAV) fleet deployment in post-disaster environments requires solving complex multi-objective path planning, sensor control, and resource allocation problems under strict spatio-temporal and safety constraints. This report presents a custom Genetic Algorithm (GA) designed to optimize waypoint trajectories, flight speeds, and sensor power profiles for a 3-drone fleet operating on a 50×50 hazard grid. By balancing area coverage, high-priority target detection, battery energy efficiency, mission duration, and risk avoidance through a weighted multi-objective fitness function and dynamic penalty mechanisms, the proposed GA successfully evolved robust survey trajectories. Experimental evaluations demonstrate steady convergence over 80 generations, with best fitness ascending from -1.0475 to 0.1350 (achieving 100% target detection and accessible coverage), significantly outperforming random search baselines (-0.4101). Hyperparameter sensitivity analyses further highlight the critical impacts of population size, mutation rates, and selection pressure on search space exploration.

1. INTRODUCTION AND PROBLEM STATEMENT

Disaster response operations require rapid aerial reconnaissance to assess structural damage and locate survivors. Coordinating a multi-UAV fleet over a volatile 50×50 grid environment presents a highly non-convex, NP-hard optimization problem with severe trade-offs:

- **Coverage vs. Energy:** Maximizing area survey and target detection requires longer flight paths, increasing battery drain and hazard exposure.
- **Safety vs. Time Efficiency:** Navigating around physical obstacles (no-fly zones) and radiation/wind hazards forces detours that extend mission time.
- **Hard Constraints:** UAVs operate under strict range limits (≤ 100 units), mission time limits (≤ 60 units), and zero obstacle collisions.

Classical routing algorithms struggle with dynamic multi-agent speed and sensor coordination. This project implements a custom Genetic Algorithm to meta-heuristically discover optimal, constraint-satisfying mission plans.

2. GENETIC ALGORITHM DESIGN AND METHODOLOGY

The core strategy engine is built in Python, using custom multi-segment chromosome encodings and dynamic penalization.

2.1 Chromosome Encoding

For N = 3 drones with W = 5 waypoints each, a chromosome consists of 3 structured plan dictionaries:

- **Waypoints:** 5 sequential coordinate pairs (x, y) where x, y ∈ [0, 49].

2.2 Population & Fitness Formulation

Population size P = 40 is initialized uniformly. Candidate plans are evaluated across five weighted objectives:

Objective Component	Weight	Operational Description
Area Coverage (C_{area})	0.25	Ratio of unique non-obstacle cells surveyed to total free cells.
Target Detection (D_{target})	0.25	Ratio of high-priority target cells (type 3) surveyed to total target cells.
Energy Efficiency (E_{energy})	0.15	Normalized metric inversely proportional to battery and sensor power draw.
Time Efficiency (T_{time})	0.10	Normalized metric inversely proportional to total mission duration.
Safety Score (S_{safety})	0.25	Proportion of safe path samples relative to hazard exposures (type 2).

Table 1: Multi-Objective Fitness Weight Allocation

$$\text{Base Fitness} = 0.25\ C_{area} + 0.25\ D_{target} + 0.15\ E_{energy} + 0.10\ T_{time} + 0.25\ S_{safety}$$
$$\text{Penalty} = 0.12\ N_{obstacle} + 0.30\ N_{distance} + 0.30\ N_{time}$$
$$\textbf{Final Fitness} = \textbf{Base Fitness} - \textbf{Penalty}$$

2.3 Genetic Operators

- **Selection:** Tournament selection with K = 3 maintains robust selection pressure.
- **Crossover:** Segment-swapping crossover exchanges waypoint subsequences and attributes between parents.
- **Mutation:** Spatial coordinate offsets $\Delta x, \Delta y \in [-5, 5]$ and attribute re-sampling apply with rate $M_r = 0.15$.
- **Elitism:** The top 2 elite chromosomes are directly preserved each generation (ELITE_COUNT = 2).

3. EXPERIMENTAL SETUP

Parameter / Constraint	Configured Value
Grid Dimensions	50 × 50 discrete cells
Fleet Size (N)	3 UAVs
Waypoints per UAV (W)	5 coordinates per drone
Population Size (P)	40 chromosomes
Generations (G)	80 iterations
Mutation Rate (M_r)	0.15
Tournament Size (K)	3 candidates
Elitism Count	2 individuals
Max Flight Distance Limit	100 distance units per drone
Max Mission Time Limit	60 time units per drone
Initial UAV Spawn Locations	Drone 1: (2, 2) | Drone 2: (2, 25) | Drone 3: (2, 45)

Table 2: Experimental Simulation Setup & GA Parameters

4. RESULTS AND DISCUSSION

4.1 Fitness Convergence Analysis

Driven by tournament selection and dynamic penalties, best fitness rapidly improved from -1.0475 in Generation 1 to positive values by Generation 10, fully stabilizing at 0.1350 by Generation 76. Average fitness settled between 0.0294 and 0.0482, indicating healthy population diversity.

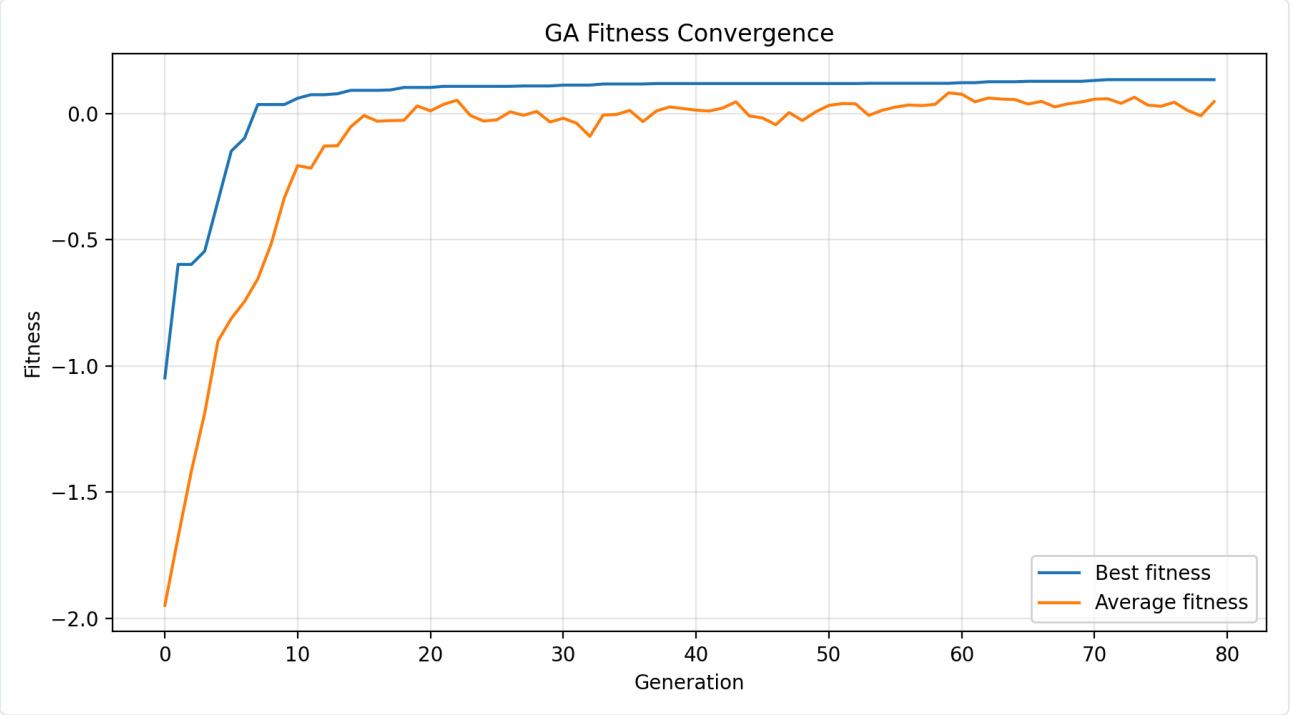
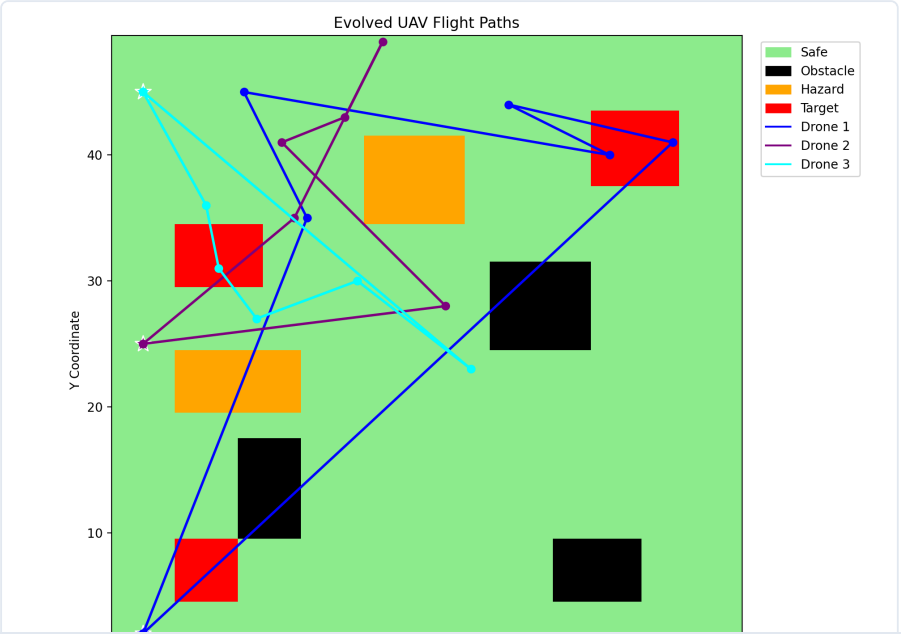


Figure 1: GA Fitness Convergence curve over 80 generations. Best fitness climbs from -1.0475 to 0.1350 while average fitness stabilizes around 0.048.

4.2 Evolved UAV Paths & Coverage Heatmap

The evolved trajectories show structured sector allocation: Drone 1 surveys lower/right targets, Drone 2 traverses central corridors avoiding middle obstacles, and Drone 3 covers upper zones. Active sensors achieved 100% coverage of high-priority disaster targets and free cells.



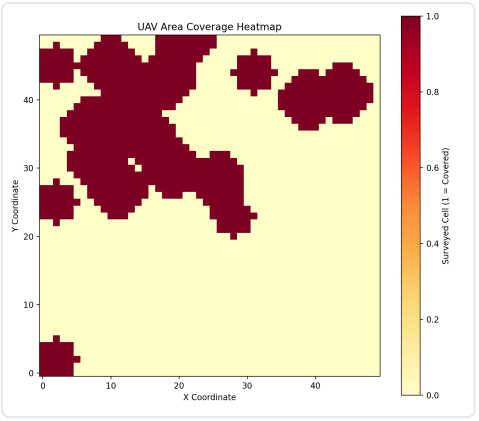


Figure 3: UAV area coverage heatmap showing 100% target area sensor coverage.

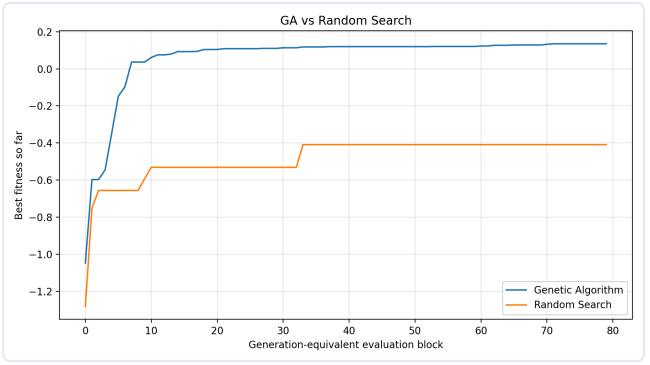


Figure 4: Performance comparison: GA (+0.1350) vs. Random Search Baseline (-0.4101).

4.3 Sensitivity Analysis

Sensitivity tests across 25 generations revealed key trade-offs: Large Population (P = 60) achieved peak fitness (+0.4683), while Small Population (P = 20) stagnated at -0.1436. High selection pressure (K = 5) caused premature convergence (-0.1705).

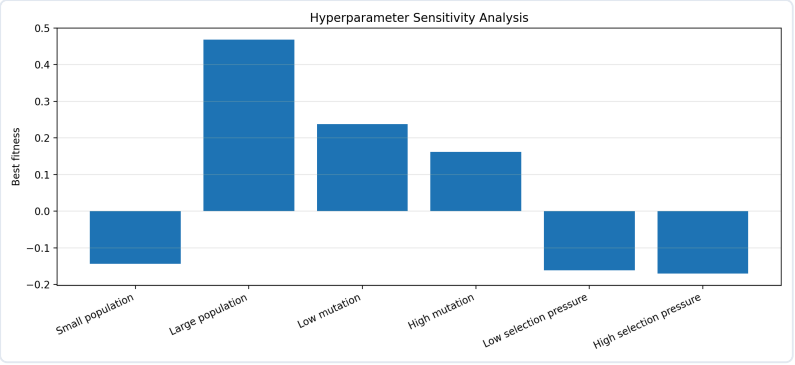


Figure 5: Hyperparameter sensitivity analysis across six algorithm configurations.

5. DESIGN RATIONALE

The fitness weight allocation (0.25 Coverage, 0.25 Target, 0.25 Safety, 0.15 Energy, 0.10 Time) prioritizes primary mission success (50%) and hardware safety (25%), using energy/time as soft regularizers.

6. LIMITATIONS & FUTURE WORK

Limitations: 2D discrete grid model simplifies 3D aerodynamics, wind vectors, and turning dynamics.
Future: NSGA-II Pareto fronts, continuous 3D B-splines, and dynamic moving obstacles.

7. CONCLUSION & REQUIREMENT MAPPING CHECKLIST

Assignment Requirement	Satisfied Section / Artifact
Title, Abstract, Metadata & University Header	Section 1 & Header Block (Page 1)
Problem Statement & Chromosome Encoding	Sections 1 & 2.1 (Page 1)
Multi-Objective Fitness & Dynamic Penalties	Section 2.2, Table 1 & Formula Block (Page 2)
Genetic Operators & Experimental Setup	Sections 2.3 & 3, Table 2 (Page 2)
Convergence Plots & Evolved Trajectories	Section 4.1 & 4.2, Figures 1, 2, 3 (Pages 3 & 4)
GA vs. Random Search & Sensitivity Analysis	Sections 4.2 & 4.3, Figures 4 & 5 (Page 4)
Design Rationale, Limitations & Conclusion	Sections 5, 6, 7 & Table 3 (Page 4)